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Impacts of rotation, tillage, cover cropping, and drainage on soil health in soybean-based cropping systems: Evidence from 4–50-year trials across the US

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ABSTRACT

Recent studies highlight conservation management practices as an effective strategy to enhance soil health. However, results vary, particularly regarding which soil health parameters respond most sensitively to these practices. More studies covering a wide range of soil types and climatic conditions are needed to assist farmers in making management decisions on production practices related to soil health. In this study, we collected soil samples (0–15 cm) from 21 (4–50 years) soybean [*Glycine max* (L.) Merr.]-based cropping systems trials across the United States (US) to assess the impact of management practices on soil health indicators. Soil indicators included wet aggregate stability (WAS), permanganate oxidizable carbon (POXC), organic matter loss-on-ignition (OM-LOI), mineralizable carbon (Min-C), water extractable organic carbon (WEOC), total organic carbon (TOC), soil extractable protein (ACE-N), total nitrogen (TN), pH, soil test phosphorus (STP), and soil test potassium (STK). Our objectives were: (i) to assess the effects of crop rotation, tillage, cover cropping, and artificial

Abbreviations: CC, continuous corn *Zea mays* L.; SS, continuous soybean; WW, continuous winter wheat *Triticum aestivum* L.; CS, 2-yr rotation sequence of corn–soybean; SC, 2-yr rotation sequence of soybean–corn; SW, 2-yr rotation sequence of soybean–wheat; GSS, grain sorghum-soybean; W/S, wheat-double crop soybean; CW/S, corn-wheat-double crop soybean; GSW/S, grain sorghum *Sorghum bicolor*-wheat-double crop soybean; SW/GS, soybean-wheat-double crop grain sorghum; CCSS, 2-yr of corn followed by 2-yr of soybean; CSW, 3-yr rotation sequence of corn–soybean–wheat; SWC, 3-yr rotation sequence of soybean–wheat-corn; CsilWS, 3-yr rotation sequence of corn silage-wheat-soybean; 1W/S-CO-PN-C, first year wheat-double crop soybean after cotton, peanut, and corn rotations; 1C, first year of corn after 5-yr of soybean; 1S, first year of soybean after 5-yr of corn; 5S, fifth year of soybean after 5-yr of corn; 5C, fifth year of corn after 5-yr of soybean; CSCOA, corn-soybean-corn-oat *Avena sativa* L.-alfalfa *Medicago sativa* L. rotation; CT, conventional tillage; NT, no-tillage; RT, reduced tillage; ST, strip tillage; CP, chisel plow; MP, Moldboard plow; WAS, wet aggregate stability; POXC, permanganate oxidizable carbon; OM-LOI, organic matter loss-on-ignition; Min-C, mineralizable carbon; WEOC, water extractable organic carbon; TC, total carbon; TOC, total organic carbon, ACE-N, soil extractable protein; Total N, total nitrogen; STP, soil test phosphorus; STK, soil test potassium.

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drainage on soil health; (ii) to inform soybean farmers about the management practices that are associated with improvements on soil health; and (iii) to develop and share a unique and open soil health dataset with the research community for future global meta-studies. To assess the effects of management practices on soil health indicators, both meta-analysis approach and linear mixed-effect models were used. Two-crop rotations were associated with greater STP values compared to a single-crop. The inclusion of cover crops was associated with greater Min-C and WEOC compared to no cover crops. No-tillage showed more acidic pH than conventional tillage. The remaining soil health indicators tested did not change in response to the management practices assessed. There were no statistically significant differences in observed soil tests between tile-drained and undrained treatments. Overall results suggest that cover crops can play an important role in building soil health in soybean-based cropping systems. Our open-access dataset provides a valuable resource for future research and meta-studies, ultimately contributing to the development of more effective management strategies for promoting more sustainable soybean cropping systems.

1. Introduction

Soil health improvement efforts to sustain ecosystem function in agricultural fields have gained attention due to more frequent drought and intense rainfall events, and the resulting concern about the soil's ability to support crop production and meet food demand. Specifically, enhancing soil health can improve ecosystem resistance and resilience by increasing the capacity of soils and agroecosystems to absorb disturbances, allowing agricultural systems to better withstand and recover from management impacts and increasing weather variability (Abdalla et al., 2019; Davis et al., 2025; Norris et al., 2020).

Soil health is generally defined as the ability of soil to serve as a vital living ecosystem that supports the life cycles of plants, animals, and humans (Lehmann et al., 2020). Due to the complexity of soil health, soil biological, physical, and chemical components are commonly used as indicators to measure or estimate 'soil health' (Maaz et al., 2023). Biological indicators, including soil organic matter (SOM), soil organic carbon (SOC), mineralizable carbon (Min-C) or soil respiration, permanganate oxidizable carbon (POXC), and extractable protein (ACE-N), are commonly utilized to assess soil health. These indicators provide insight into the levels and composition of SOM, as well as the presence and activity of soil microorganisms, and are closely associated with various soil functions and processes (Nunes et al., 2020). Physical soil properties, such as aggregate stability, are valuable indicators of soil health due to their association with water infiltration and storage, the formation of habitats for microorganisms and roots, and the supply of essential plant nutrients (Rieke et al., 2022). Chemical or soil fertility analytical tests, including soil pH and available nutrients, are critical because soil chemical conditions regulate the availability and abundance of essential nutrients required for crop growth, which might impact overall yield, as well as for their impact on microbial communities (Fierer, 2017; Norris et al., 2020). Generally, biological, physical, and chemical indicators are selected based on their relevance, consistency, cost-effectiveness, and sensitivity to management practices (Morrow et al., 2016). Various other indicators have been suggested by researchers, and those considered important, along with their corresponding response levels, have often varied by location.

Conservation management practices, including minimal or no-tillage, diverse crop rotations with multiple crops, and cover crop adoption are generally linked to maintaining and increasing soil health parameters by enhancing soil biodiversity, improving soil water and nutrient cycling, and reducing soil compaction and erosion (Chahal et al., 2021; Farmaha et al., 2022; Nunes et al., 2018; Qin et al., 2023). Cover crops are also known for reducing nitrogen (N) leaching and weed pressure (Abdalla et al., 2019; Osipitan et al., 2018). Diversified crop rotations can reduce pest (weed, pathogen, and insect) pressure, increase drought tolerance, as well as improve crop yield when compared to monoculture systems (Agomoh et al., 2020; Liu et al., 2022). The adoption of conservation management practices, such as cover cropping is increasing in the US. For example, the total cropland area planted with cover crops in 2022 (7.2 million hectares) was approximately 17 % higher than in 2017 (USDA-NASS, 2022). Cover crops have been positively associated with soil health carbon indicators, such as Min-C, organic matter loss-on-ignition (OM-LOI), and SOC (Liptzin et al., 2022; Wood and Bowman, 2021; Wulanningtyas et al., 2021).

Artificial drainage is a management practice used to alleviate waterlogged or saturated soils by draining the excess water from the soil profile. Drainage can improve soil aeration in the plant root zone, allow for earlier planting due to more suitable days for fieldwork in early spring, and increase crop productivity compared to poorly-drained soils (Mourtzinis et al., 2021). However, this practice has been associated with a decrease in SOM (Fernández et al., 2017; Kumar et al., 2014). In general, water-logged soils lead to reduced oxygen and microbial activity levels, resulting in slower organic matter decomposition and more organic carbon (OC) accumulation in the soil compared to drained soils (Fernández et al., 2017; Richardson et al., 2023). While artificial drainage may increase crop yield, its impact on soil health needs further investigation, particularly in regions with extensive use like the North Central US (USDA-NASS, 2022). Although wetland ecosystems can accumulate high SOM under saturated conditions (Ji et al., 2020), we focus on drained agricultural soils, where aeration accelerates SOM decomposition and likely affects agronomic soil health.

Studies exploring the effects of management practices on soil health indicators have reported divergent results regarding which management practices are positively or negatively associated with soil health (Agomoh et al., 2020; Chamberlain et al., 2020; Congreves et al., 2015). This could be due to a range of factors such as soil type, climate, region, management practice interactions, study duration, and soil sampling depth (Agyei et al., 2024; Nunes et al., 2020). Therefore, experiments that cover a variety of different soils and climate regions that focus on conservation management practices to maintain or improve soil health are needed to better understand the impact of these practices on soil biological, chemical, and physical indicators. Existing field experiments that have been actively maintained for multiple years present an opportunity to explore whether conservation practices enhance soil health, as changes in SOM and related indicators typically occur slowly in response to management.

Given the existing literature and the observed benefits of these practices, we hypothesized that conservation management practices—-specifically the adoption of cover crops, diversified crop rotations (two or three crops), and reduced tillage (no-tillage or minimal tillage)—would enhance soil health parameters in soybean [*Glycine max* (L.) Merr.] cropping system trials across the US. Conversely, we also anticipated that artificial drainage, known for its potential to decrease soil organic matter, would be negatively associated with these soil health indicators. Building soil health in soybean-based systems can be particularly challenging due to factors such as limited crop diversity often associated with soybean rotations, frequent tillage practices that disrupt soil structure, nutrient imbalances resulting from soybean's nitrogen-fixing but nutrient-depleting nature, and increased pest and disease pressures. These challenges may slow improvements in soil biological activity, organic matter accumulation, and overall soil function compared to more diverse or less intensively managed cropping systems (Dietz et al., 2024).

We investigated whether crop rotation, cover crops, tillage, and artificial drainage were associated with changes in soil test indicator values (soil health parameters and soil chemical and physical properties). This study was approached through the lens of soybean cropping systems for several reasons, including the historical focus on corn in soil health research, added challenges for building soil health in soybean-based systems, and collaborative momentum in the soybean research community. The specific objectives of our study were: i) to use data measured from 21 (4–50 years) field experiments across the US to explore the relationship between management practices and soil health indicators; ii) to inform soybean farmers about the management practices that are associated with improvements on soil health; and iii) to generate a unique and open dataset for soil health assessment for meta-analyses.

2. Materials and methods

2.1. Study description

This study comprised soil sample collection from 21 (4–50 years) experimental agricultural research trials (Fig. 1). Sites included in this study had experimental treatments arranged in a randomized complete block design, with 3 or 4 replications, that compared the effects of at least one of four practices: cover cropping, crop rotation, tillage, and artificial drainage. All plots were managed similarly prior to the establishment of each study, ensuring that observed differences among treatments reflect the effects of management rather than differences in initial conditions. Agronomic management and field histories for all sites were documented (Table 1) by each collaborator of this project.

2.2. Field management

None of the sampled fields have received manure application nor other biomass additions since their establishment. Study 10 in Hickory Corners, MI received lower different rates of phosphorus (P) and potassium (K) among treatments, whereas all treatments in the remaining trials received the same rates of P and K. The drainage sites in Wells, MN (study 11) and Fargo, ND (study 15) were rainfed, with no supplementary irrigation applied. Both studies aimed to compare the effects of artificial drainage versus no drainage. The treatments involved tile drainage systems with control structures: in the drained plots, the control structures were fully open, while in the undrained plots, they were closed. Both the tile-drained and undrained plots were managed under conventional and reduced tillage systems.

2.3. Soil sampling

A total of 625 plots were sampled from the 21 trials in 2023. It is important to note that this study does not assess changes in soil health indicators over time, but rather evaluates their current status in relation to differences in management practices. Soil samples were collected in fields from mid-April to mid-June, near or at crop planting according to each location's planting crop schedule. For chemical and biological analysis, 20 sub-samples at 15 cm depth were collected from each plot with a 2.5 cm diameter soil probe. One sample (20 composited 15 cm sub-samples) from each blocked replication was analyzed for soil texture (sand, silt, and clay).

For wet aggregate stability (WAS) soil analysis, a sharpshooter shovel was used to collect 3 sub-samples per plot. Briefly, the shovel was used to vertically cut into the soil (at a depth of 20 cm) at a 90° angle

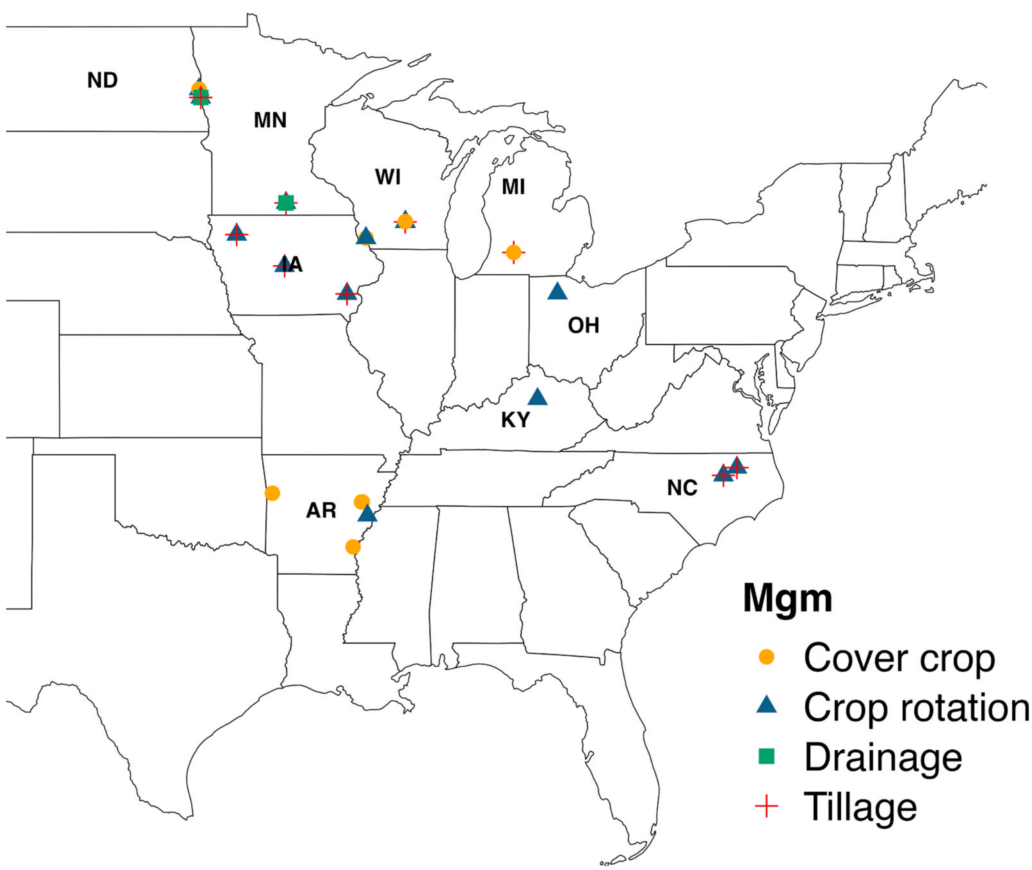

Fig. 1. Map of 21 long-term experiments included in the study. Field trials that included more than one management (Mgm) practice are represented by the different colors and shapes in the figure. Number of trials in each state: Arkansas-AR (5), Iowa-IA (3), Kentucky-KY (1), Michigan-MI (1), Minnesota-MN (1), North Carolina-NC (2), North Dakota-ND (2), Ohio-OH (1), Wisconsin-WI (5).

Table 1
Characteristics of the 21 studies selected to evaluate soil health indicators response to management practices. The meta-analysis column indicates the management practices included in this analysis when treatment (cover crop, two crops in rotation, three crops in rotation, no-tillage, and minimal tillage) and control (no cover crop, single-crop, conventional tillage) interventions were present in the same trial.

Study number	Study name	State-location	Year Established	Crop rotation	Tillage	Annual irrigation	Sand (%)	Silt (%)	Clay (%)	Soil order	Texture class	Meta-analysis treatments[c]
1	Soybean Cover Crop Kibler	AR-Kibler[d]	2017	SS	NT	yes	22	42	36	Mollisol	silt loam	cover crop
2	Soybean Cover Crop Rohwer	AR-Rohwer[d]	2017	SS	NT	yes	23	62	15	Alfisol	silt loam	cover crop
3	Soybean Cover Crop Colt	AR-Colt[d]	2017	SS	NT	yes	16	62	22	Alfisol	silt loam	cover crop
4	Soybean/ Corn Cover Crop Colt	AR-Colt[d]	2016	CS	NT	yes	9	64	27	Alfisol	silt loam	cover crop
5	G11 Rotation	AR-Marianna	2013	CC; CS; SS; W/S; GSS; CW/S; GSW/S; SW/ GS	CT	yes	10	68	22	Alfisol	silt loam	two crops in rotation; three crops in rotation
6	Long-term tillage by rotation	IA-Sutherland	2005	CC; CS	NT; CP; MP	no	10	45	45	Mollisol	silty clay loam	two crops in rotation; no-tillage
7	Long-term tillage by rotation	IA-Ames	2005	CC; CS	NT; CP; MP	no	37	28	35	Mollisol	clay loam	two crops in rotation; no-tillage
8	Long-term tillage by rotation	IA-Crawfordsville	2005	CC; CS	NT; CP; MP	no	8	51	41	Mollisol	silty clay loam	two crops in rotation; no-tillage
9	Grove Long-term rotation trial	KY-Lexington	1986[a] & 2008[b]	CC; CW/S; CCSS	NT	no	11	55	34	Alfisol	silt loam	two crops in rotation; three crops in rotation
10	LTER Project	MI-Hickory Corners[e]	1989	CSW	CT; NT; RT; CP	no	45	33	22	Alfisol	loam	no-tillage; minimal tillage; cover crop
11	Drainage-Tillage Wells	MN-Wells	2012	CS; SC	CT; NT	no	31	33	36	Mollisol	silty clay loam	no-tillage
12	Lewiston Tillage Rotation	NC-Lewiston	1999	1 W/S-CO-PN-C	CT; ST	no	76	9	15	Ultisol	sandy loam	minimal tillage
13	Rocky Mount Tillage Rotation	NC-Rocky Mountain	2004	1 W/S-CO-PN-C	CT; ST	no	53	32	15	Ultisol	sandy loam	minimal tillage
14	Bell Family Farms NW−37	ND-Gardner[f]	2017	SWC	RT	no	14	27	59	Vertisol	silty clay	cover crop
15	Fargo NW 22	ND-Fargo	2008	SW	CT, RT	no	14	26	60	Vertisol	silty clay	minimal tillage
16	Rotation Study	OH-Custar	2012	CS; CSW	NT	no	26	26	48	Alfisol	silty clay loam	no control intervention to include in the meta-analysis
17	Long term corn/soy rotation	WI-Arlington	1983	CC; CS; SS; 1 C; 5 C; 1S; 5S	CT; NT	no	17	53	30	Mollisol	silt loam	two crops in rotation; no-tillage
18	Long term corn/soy/ wheat rotation	WI-Arlington	2002	CC; SC; CWS; CSW; WW; CsilWS	NT	no	18	52	30	Mollisol	silt loam	two crops in rotation; three crops in rotation
19	Arlington System I	WI-Arlington[g]	2019	CS	CT; NT	no	15	56	29	Mollisol	silt loam	no-tillage; cover crop
20	Lancaster System II	WI-Lancaster[g]	2019	CS	CT; NT	no	17	58	25	Alfisol	silt loam	no-tillage; cover crop
21	Long Term Crop Rotation	WI-Lancaster	1966	CSW; CSCOA	NT	no	15	62	23	Alfisol	silt loam	no control intervention to include in the meta-analysis

Abbreviations: CC, continuous corn (*Zea mays* L.); SS, continuous soybean; WW, continuous winter wheat (*Triticum aestivum* L.); CS, 2-yr rotation sequence of corn–soybean; SC, 2-yr rotation sequence of soybean–corn; SW, 2-yr rotation sequence of soybean–wheat; GSS, grain sorghum-soybean; W/S, wheat-double crop soybean; CW/S, corn-wheat-double crop soybean; GSW/S, grain sorghum (*Sorghum bicolor*)-wheat-double crop soybean; SW/GS, soybean-wheat-double crop grain sorghum; CCSS, 2-yr of corn followed by 2-yr of soybean; CSW, 3-yr rotation sequence of corn–soybean–wheat; SWC, 3-yr rotation sequence of soybean–wheat-corn; CsilWS, 3-yr rotation sequence of corn silage-wheat-soybean; 1 W/S-CO-PN-C, first year wheat-double crop soybean after cotton, peanut, and corn rotations; 1 C, first year of corn after 5-yr of soybean; 1S, first year of soybean after 5-yr of corn; 5S, fifth year of soybean after 5-yr of corn; 5 C, fifth year of corn after 5-yr of soybean; CSCOA, corn-soybean-corn-oat (*Avena sativa* L.)-alfalfa (*Medicago sativa* L.) rotation; CT, conventional tillage; NT, no-tillage; RT, reduced tillage; ST, strip tillage; CP, chisel plow; MP, Moldboard plow; [a]CC and CW/S rotation was started in 1986; [b]CCSS rotation was started in 2008. [c]Meta analysis management treatments were compared to their respective controls (e.g., cover crop vs. no cover crop, two crops in rotation vs. single-crop, three crops in rotation vs. single-crop, no-tillage vs. conventional, minimal tillage vs. conventional. AR, Arkansas; IA, Iowa; KY, Kentucky; MI, Michigan; MN, Minnesota; NC, North Carolina; ND, North Dakota; OH, Ohio; WI, Wisconsin. All trials were managed according to the corresponding University guidelines. The sites with cover crops included individual cover crop species as well as mixed cover crops. [d]Studies 1–4 included individual cereal rye (*Secale cereale* L.), black-seeded oat (*Avena sativa* L.), barley (Hordeum vulgare L.), hairy vetch (*Vicia villosa* Roth), and Austrian winter pea [*Pisum sativum var. arvense* (L.) Poir] and combinations of seven-top turnip (*Brassica rapa* L.), cereal rye, and crimson clover (*Trifolium incarnatum* L.), and a mix of black oats and Austrian winter pea. [e]Study 10 in Hickory, MI had red clover (*Trifolium pratense* L.). [f]Study 14 in Gardner, ND included winter wheat, cereal rye, and camelina [*Camelina sativa* (L.) Crantz], and [g]studies 19 and 20 in Arlington and Lancaster, WI had cereal rye.

four times, creating a square. From one of the vertical sides of the hole, a slice of soil measuring 15-cm in depth, 5-cm in width, and 2-cm in thickness was collected. The slices of soil (3 sub-samples per plot) were then gently broken up by hand and carefully mixed to minimize disruption of natural aggregation. Roughly 250 mL of the composite sample was placed in a plastic container for further analysis. Samples were stored in a refrigerator or cold room at 4 °C upon returning from the field if they were not immediately sent to the laboratory.

2.4. Soil laboratory analysis

Soil health indicators were carefully selected based on previous literature, tests recommended by the Soil Health Institute (SHI; https://soilhealthinstitute.org/), and tests used in current tools available to measure soil health such as the Comprehensive Assessment of Soil Health [CASH] (Gugino et al., 2009) that rely on a specific set of measures of soil biological, physical, and chemical components. These are also tests about which we receive questions from farmers.

All collected soil samples were analyzed at Regen Ag Lab, LLC, in Pleasanton, NE for all indicators assessed in this study. The POXC was measured following the method described by Weil et al. (2003). Briefly, 2.5 g of air-dried, ground soil passed through a 2 mm mesh sieve was extracted with 18 mL of deionized H_2O and 2 mL of potassium permanganate. The mixture was shaken for 2 min, then allowed to rest for an additional 8 min. Subsequently, 0.5 mL of the extractant was diluted in 49.5 mL of deionized H_2O to halt further oxidation. The samples were then analyzed using a spectrophotometer at 550 nm. The POXC is interpreted as a moderately processed fraction of SOC, based on the color change of potassium permanganate as it is reduced by compounds in the soil (Culman et al., 2012), and should not be interpreted as "active carbon" (Woodings and Margenot, 2023). The Min-C or soil respiration was measured by the cumulative CO_2 evolved after a 24 h (±1 h) incubation at 24°C (Franzluebbers et al., 1996). Air-dried, sieved soil (40 g) was incubated in sealed glass jars containing 17 mL of deionized H_2O and analyzed using a Soil Respiration-1 (SR-1) infrared gas analyzer. Min-C represents the size of the pool of readily available OC and the activity level of microbial communities in the soil.

The WAS was assessed by the fraction of aggregates that remain intact after exposure to destabilizing stressors (Mikha and Rice, 2004). A 4.0 g sample of air-dried and ground soil was placed onto the 0.25 mm sieve (to collect macroaggregates fractions of the soil, > 0.25 mm) on an automated sieving machine, with deionized H_2O cans positioned below. The machine was set to 34 strokes per min for 3 min. After sieving, the deionized water can was removed and replaced with a can of sodium hexametaphosphate. Aggregates remaining on the sieve were submerged and crushed until all aggregates passed through the sieve. The can was then placed in the oven (110 °C) overnight to dry and then reweighed. The 0.25 mm sieve was replaced with a 0.053 mm sieve to collect microaggregates (< 0.25 mm and > 0.053 mm). The total WAS percentage present in the soil sample was calculated.

The percentage of SOM was quantified by the loss-on-ignition (LOI) method. Soil samples were air-dried, ground, and sieved through a 2 mm sieve prior to analysis. This involved heating a 2.0 g soil sample in a muffle furnace at 375 °C for 2 h and measuring the change in mass (Combs and Nathan, 1998). For water extractable organic carbon (WEOC) content, soil was air-dried, ground, and passed through a 2 mm mesh sieve. Then, 2.0 g of soil was shaken with deionized H_2O in a 50 mL centrifuge tube for 5 min at 3000 rpm. The supernatant was then filtered, and OC was analyzed immediately using an elemental analyzer. Total carbon (TC), and total nitrogen (TN) contents were determined by combustion analysis using a LECO CN analyzer (LECO Corporation, St. Joseph, MI). Briefly, soil was air-dried and ground to pass through a 2 mm mesh screen and 1.0 g of sample was weighed into a combustion boat and introduced into an oxygen-rich environment at 1350 °C (Gavlak et al., 1996). The resulting gasses were analyzed to quantify total C and N content. To quantify soil total organic carbon (TOC), a 1.0 g air-dried, ground, and sieved soil sample was treated with sulfurous acid to remove carbonate minerals. The remaining organic material was then combusted entirely using a LECO analyzer to determine its TOC content (Nelson and Sommers, 1996).

The ACE-N was quantified following Hurisso et al. (2016). Briefly, 3.0 g of air-dried, ground soil passed through a 2 mm mesh sieve were shaken with a neutral citrate buffer to facilitate disaggregation and protein extraction via autoclave treatment. The extracted protein concentration was then determined colorimetrically using a standard curve at 562 nm with a microplate spectrophotometer. Soil water pH was measured with a glass electrode in a 1:1 soil/water suspension (Peters et al., 2012). Soil test phosphorus (STP) was measured using the Mehlich-3 extractant and quantified with a Lachat flow injection analyzer (Frank et al., 1998). Soil test potassium (STK) was quantified by shaking 2.0 g of soil with ammonium acetate for 5 min at 240 rpm (Warncke and Brown, 1998). The supernatant was then filtered and analyzed using an iCAP 6500 Duo ICP-OES (Thermo Fisher Scientific, Waltham MA, US). Soil samples for pH, STP, and STK were air-dried, ground, and passed through a 2 mm mesh sieve prior to analysis. Although chemical soil health indicators are highly dependent on inherent soil properties and management inputs, these indicators are known to be responsive to management practices (Nunes et al., 2020) and therefore were included in our soil health dataset.

2.5. Statistical analysis

All statistical analyses were performed in R Studio version 4.4.0 (R Core Team, 2024). Correlations between the soil health indicators were assessed for the full dataset (n = 625) using Pearson's correlation coefficients ($P < 0.05$) and distributions of the soil health indicators were obtained with histograms using the "ggpairs" function in the *GGally* package (Schloerke et al., 2024).

A meta-analysis was conducted, following the methodology of Liptzin et al. (2022), to examine the response of soil health indicators to various management practices—including crop rotation, minimal tillage, no-tillage, and cover crops (Table 2). The analysis was performed using the "rma" function in the metafor package (Viechtbauer, 2010). Meta-analysis is a valuable technique to assess the effect of management treatments on soil health indicators across sites with varying climates and soil conditions (Eagle et al., 2017; Gurevitch et al., 2018). This approach reduces uncertainty by pooling observational data from multiple field experiments or data sources, resulting in a single effect size that represents the response to a given management practice (Young et al., 2021). The effect size was predicted as the log-transformed ratio of means between pairs of treatments within sites that differed in one management practice (Table 2). Individual models were developed for each soil health indicator, comparing the log response ratio between treatment and control (Table 2), considering site as a random effect. Only studies that comprised the appropriate treatment pairs were included in the meta-analysis (Table 1). At some sites, more than one

Table 2
Description of management practices for the studies included in the meta-analysis.

Management practices	Treatment description	Control	Paired observations	Trial-level occurrences[a]
Two crops in rotation	Crop rotation with two crop types in conventional and no-tillage systems	Single-crop monoculture in conventional and no-tillage systems	199	7
Three crops in rotation	Crop rotation with three or more crops in conventional and no-tillage systems	Single-crop monoculture in conventional and no-tillage systems	46	3
Minimal tillage	Lower tillage intensity such as reduced tillage, and strip tillage with two or more crops in rotation	Conventional tillage including any higher tillage intensity such as conventional tillage, chisel plow, deep ripper, and moldboard plow with two or more crops in rotation	121	4
No-tillage	No-tillage with a single or two or more crops in rotation	Conventional tillage including any higher tillage intensity such as conventional tillage, chisel plow, deep ripper, and moldboard plow with a single or two or more crops in rotation	208	8
Cover crops	Cover crop in no-tillage or reduced tillage systems with a single or two or more crops in rotation	No cover crop in no-tillage or reduced tillage system with a single or two or more crops in rotation	162	8

Total number of studies included in the meta-analysis (19). [a]At some experimental trials more than one management treatment was present (e.g. two crops in rotation and no-tillage).

treatment pair was present (e.g. continuous corn with conventional tillage vs no-tillage and corn/soybean with conventional tillage vs no-tillage). Study 10 in Hickory, MI was excluded from STP and STK comparisons because it included different P and K rates among the treatments.

Artificial drainage treatments were only present in two sites, study numbers 11 and 15 (Wells, MN and Fargo, ND, respectively). Because of this low site number, artificial drainage was not included in the meta-analysis. Instead, we analyzed these two studies as a case study with linear mixed-effect models using the function "lmer" of the *lme4* package (Bates et al., 2015). Each soil health indicator response variable was fitted in a separate model considering drainage as fixed effect and tillage and replication nested within the site as random effects. Assumptions of normality and homogeneity of variance of each model were checked by visually examining the residuals. All fitted models were evaluated via ANOVA Type III Wald chi-square tests using "anova" function (*car* package; Fox and Weisberg, 2019).

3. Results

3.1. Association between soil health indicators

Correlation analysis was used to examine the relationships between soil health indicators (Fig. 2). Out of 66 pairs, including biological (POXC, OM-LOI, MIN-C, WEOC, ACE-N, TOC, and TC), physical (WAS), and chemical (TN, pH, STP, and STK) soil health indicators, 60 were significantly correlated to each other ($p \leq 0.05$). Moderate to strong positive correlations were observed between biological indicators ($r \geq 0.57$). The WAS showed weak ($r = 0.31$; $P \leq 0.001$) to moderate correlation ($r = 0.60$; $P \leq 0.001$) with biological and chemical indicators, respectively. Weak to strong correlations were observed between chemical indicators and biological indicators. The strongest correlation between TOC and TC ($r = 1.00$; $P \leq 0.001$) indicates that OC represents the predominant component of the TC content for the soils assessed in this study. Based on this finding, TC was excluded from further analysis, as the TOC measurement offers a more targeted assessment of SOM content.

3.2. Changes in soil health indicators due to management practices

Meta-analysis results showed that the soil health indicators were associated with some management practices. Three-crop rotations were not associated with changes in any of the soil health indicators compared to a single-crop (Fig. 3). On the other hand, two-crop rotations were associated with greater STP, with ratio of means estimate of 0.12 compared to a single-crop ($P = 0.04$). Cover cropping was associated with greater Min-C and WEOC compared to no cover crop ($P \leq 0.05$). The 0.18 Min-C mean ratio estimate corresponds to approximately 18 % greater Min-C for the cover crop treatment relative to the no cover crop control. Similarly, the WEOC estimate of 0.07 suggests a slight positive effect, with a 7 % greater WEOC for the cover crop treatment compared to the no cover crop. No-tillage was associated with lower pH, with ratio of means estimate of -0.02 compared to conventional tillage ($P = 0.002$). The conservation management practices assessed (no-tillage, minimal tillage, cover crops, and two- and three-crop rotations) were not associated with significant changes in POXC, WAS, OM-LOI, TOC, ACE-N, TN, and STK indicators. Soil health indicators were not significantly different between tile-drained and undrained treatments ($P > 0.05$; Fig. 4).

4. Discussion

4.1. Association between management practices and soil health indicators

Our dataset presents a snapshot of the associations between management practices and soil health indicators using 0–15 cm soil samples from soybean cropping system trials across the US (ranging from 4 to 50 years in duration). The meta-analysis revealed that three-crop rotations compared to single-crop systems did not result in greater soil health indicator values. This suggests that while rotation diversity may play a role in agricultural practices, its effect on the soil health indicators assessed in this study may not be as pronounced as initially expected. More diverse crop rotations with soybean (two or more crops in rotation) did not enhance biological soil health indicators when compared to monoculture corn, soybean, and wheat systems. This outcome could be due to the need for specific crop combinations (Nunes et al., 2018; King and Blesh, 2018) or due to other management practices. Richardson et al. (2023) found that continuous corn and rotations with alfalfa (*Medicago sativa* L.), which produced greater biomass, resulted in greater soil C and N. Their study involved 218 operating farm fields across Wisconsin and southern Minnesota with at least the last 5 years of rotation history. Soybean returns minimal crop residue with a low C to N

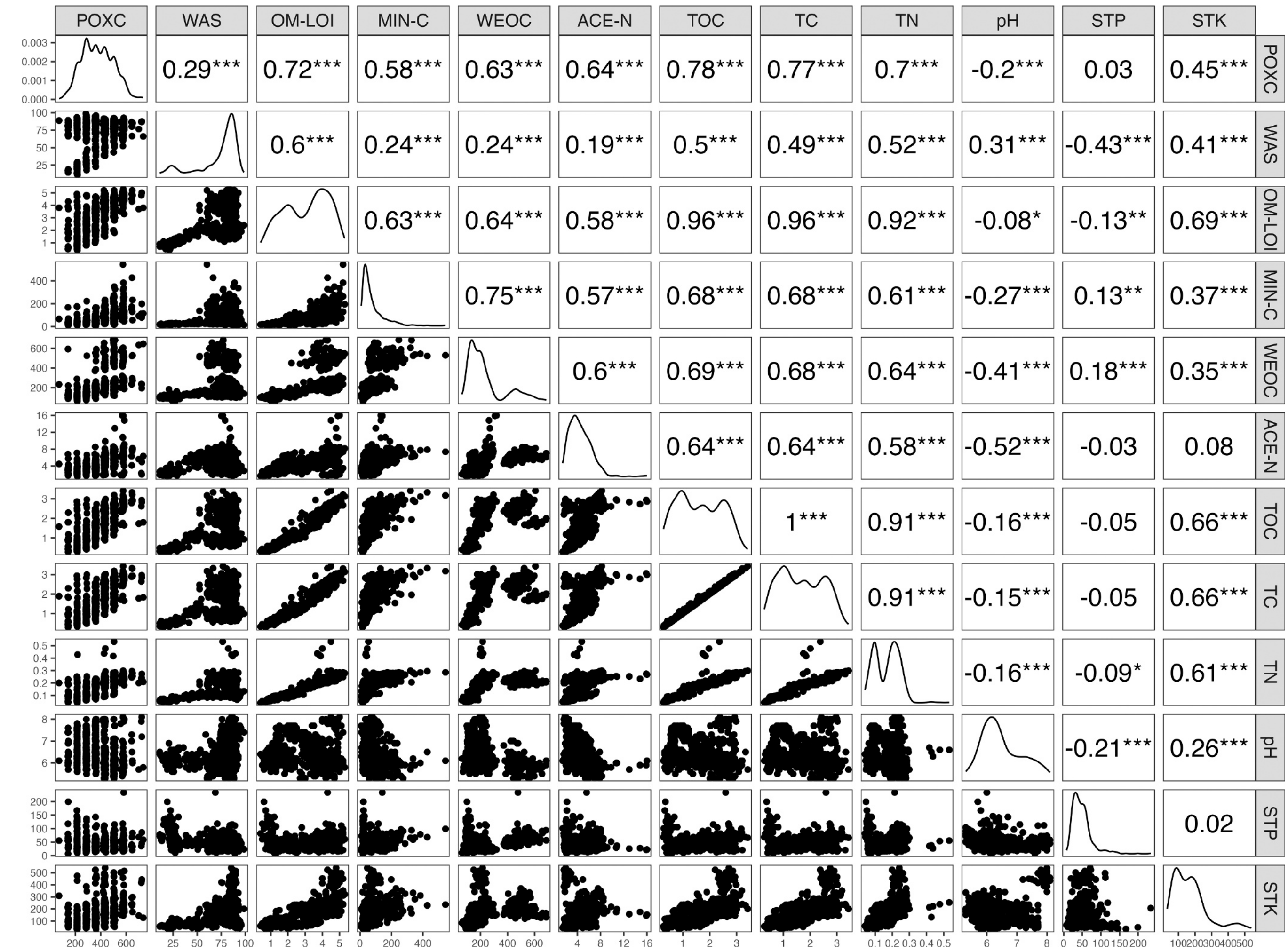

Fig. 2. Soil health indicators correlation matrix among soil health indicators across all sampled long-term experiment plots (n = 625). The scatter plots on the bottom left correspond to Pearson's correlation coefficients (r) at the top right. The diagonal density plots show distributions of the soil health indicators. Correlation values with *, **, and *** are significant at the P < 0.05, $P < 0.01$, and $P < 0.001$ probability level, respectively. WAS is wet aggregate stability, POXC is permanganate oxidizable carbon, OM-LOI is organic matter loss-on-ignition, Min-C is mineralizable carbon, WEOC is water extractable organic carbon, TC is total carbon, TOC is total organic carbon, ACE-N is soil protein, TN is total nitrogen, STP is soil test phosphorus, STK is soil test potassium.

ratio to the system after harvest, leading to a lower C to N ratio in the soil compared to cereal grains like corn, wheat, and sorghum (Agomoh et al., 2021; Chamberlain et al., 2020), which might have contributed to no differences in the biological soil health indicators between crop rotations.

It is important to note that in our study, comparisons involving three crops in rotation exhibited wider confidence intervals. This may be attributed to the limited number of studies and observations, with one study on three-crop rotation managed under a no-tillage system (n = 26) and two studies under a conventional tillage system (n = 20), limiting our statistical power. A difference in STP in rotations involving two crops compared to single-crop systems was observed. This finding aligns with previous research highlighting the potential of diversified rotations to enhance soil nutrient status (Condron et al., 2013). Crop sequences involving legumes and cereals in alternation can improve soil P resources and promote efficient crop P utilization compared to monoculture practices (Łukowiak et al., 2016). In our dataset, 70 % of the single-crop plots consisted of continuous corn, whereas 94 % of the two-crop rotations were corn-soybean rotations. Consequently, it is likely that the two-crop rotation would remove less P and have greater STP compared to the single-crop treatment. Monoculture of corn is known for higher P removal rates at harvest compared to other crops, such as soybean and wheat (Culman et al., 2019; Giannini et al., 2024). Additionally, our study design considered both single-crop and two-or-more-crop rotations under conventional and no-tillage systems. Tillage practices have also been linked to P accumulation in soil, with no-tillage promoting greater P accumulation in topsoil (0–10 cm) compared to conventional tillage (Rodrigues et al., 2016). However, when comparing tillage treatments, our data did not reveal any significant differences in STP. It is important to note that the specific effects of crop rotation on STP can be influenced by several factors, including soil type, climatic conditions, and long-term fertilizer management practices (Margenot et al., 2024).

In the present study, the inclusion of cover crops in continuous soybean and corn-soybean rotations was associated with greater levels of Min-C and WEOC. This result might be associated to the carbon input from cover crops to the soil, leading to greater Min-C and WEOC levels. These active organic matter pools are more sensitive to management practices because they consist of rapidly cycling organic material, unlike much of the bulk SOM, which decomposes more slowly (Hurisso et al., 2016). Min-C and WEOC were also moderately correlated (r = 0.75; Fig. 2), suggesting that a portion of the WEOC pool serves as a readily available carbon source for microbial activity. Min-C, and WEOC have previously shown to be more sensitive to management practices (Hurisso et al., 2018; Nyabami et al., 2024) and moderate correlated to each other (r > 0.5; Liptzin et al., 2022). Other studies have also

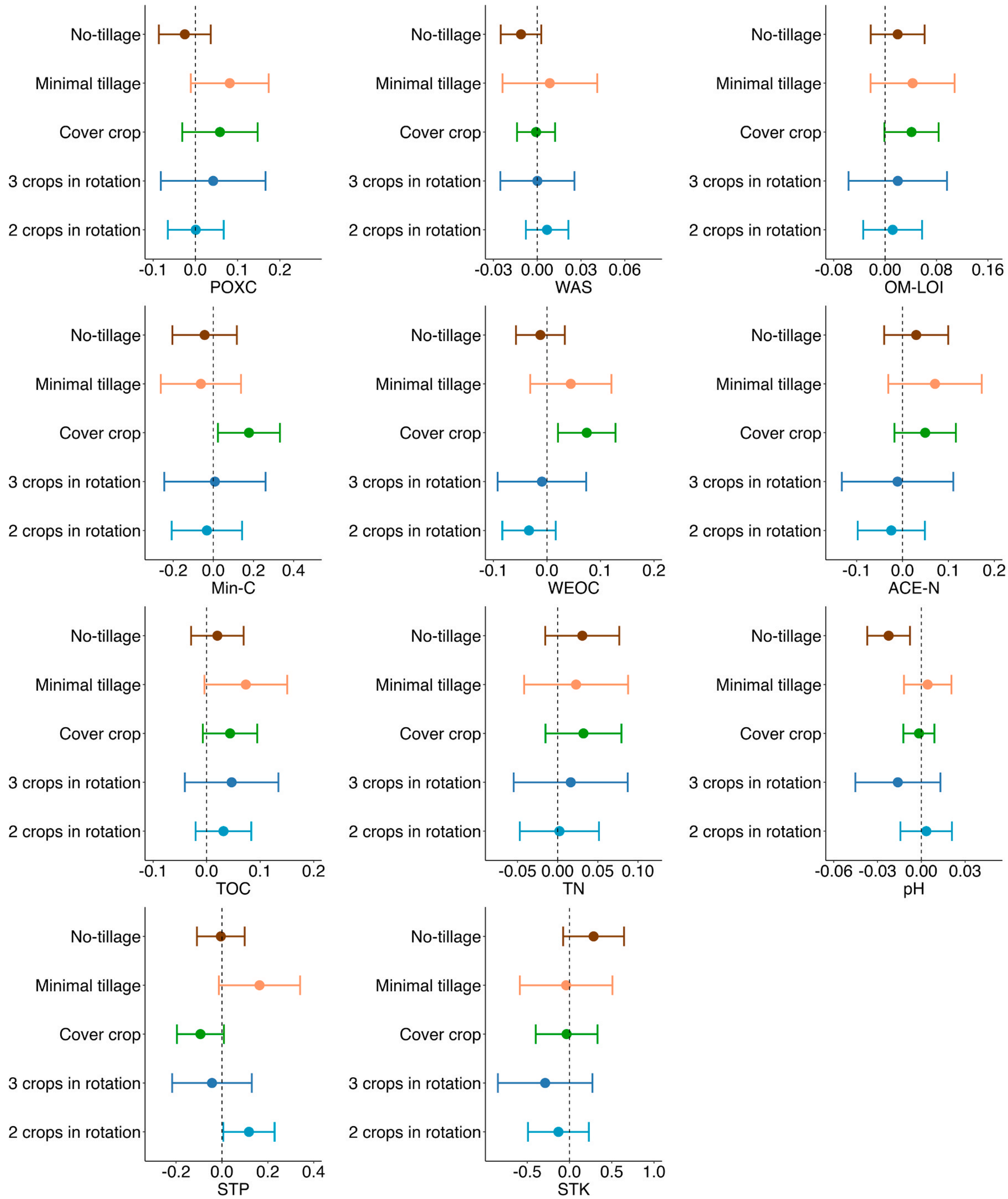

Fig. 3. Log response ratios of management practices for each soil health indicator. Effect sizes are expressed as the natural logarithm (ln) of the ratio between treatment and control means (i.e., ln[cover crop/no cover crop]). Positive values indicate increased indicator values in the treatment relative to the control, while negative values indicate reductions. The vertical black dashed line at zero represents no effect. Dots represent mean effect sizes, and whiskers show 95 % confidence intervals. Effect sizes are considered statistically significant when the confidence interval does not include zero. Table 2 provides details on the control treatment for each management practice. WAS = wet aggregate stability; POXC = permanganate oxidizable carbon; OM-LOI = organic matter loss-on-ignition; Min-C = mineralizable carbon; WEOC = water extractable organic carbon; TOC = total organic carbon; ACE-N = soil protein; TN = total nitrogen; STP = soil test phosphorus; STK = soil test potassium.

indicated similar findings. Wood and Bowman (2021), exploring 5 years of cover crop use on 78 farm-led corn trials across 9 US states, found that cover cropping was associated with positive changes in POXC, Min-C, and OM-LOI. Unlike our results, Wood and Bowman (2021) also reported a positive impact of cover crops on WAS. A global meta-analysis reported that the addition of cover crops in rotations with corn, soybean, sorghum, wheat, and other crops increased TC by 8.5 % compared to rotations without cover crops (McDaniel et al., 2014). Liptzin et al. (2022) also reported that cover crops are positively associated with Min-C and WEOC compared to no cover crops using data from 10 cover crop long-term experimental research trials.

Our study also showed more acidic soil pH in no-tillage compared to conventional tillage. This reduction is likely attributed to the higher residue accumulation and increased organic matter at or near the soil surface in no-tillage systems, which leads to soil acidification over time via microbial decomposition processes (Kumar et al., 2012; Li et al.,

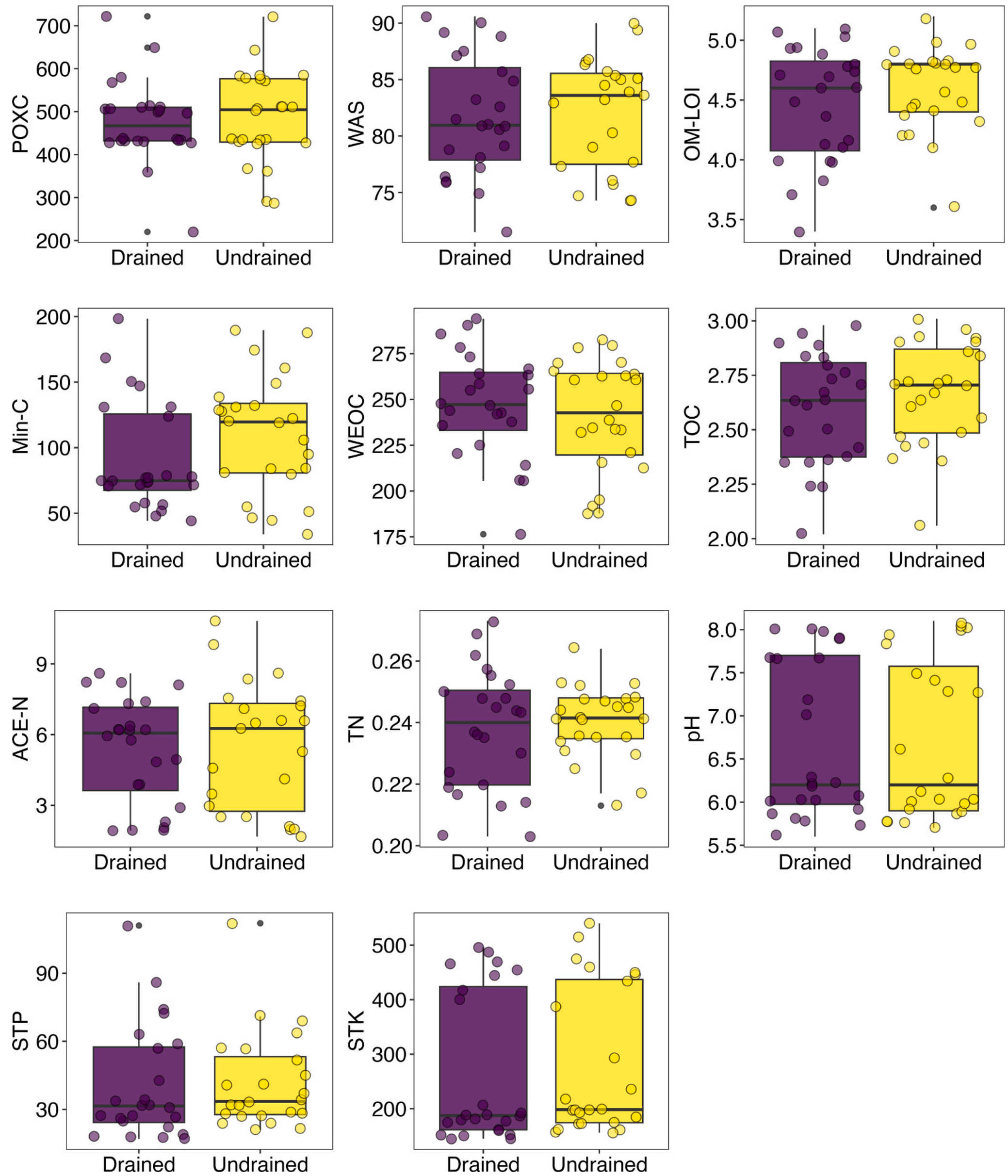

Fig. 4. Boxplots with jittered points for soil health indicators in response to tile-drained and undrained treatments (n = 24). Boxes represent the interquartile range (25th to 75th percentile), and the solid black line inside each box denotes the median. Whiskers extend to the most extreme values within 1.5 times the interquartile range; values beyond this range are shown as individual points (outliers). Jittered points are colored by drainage treatment and represent individual observations. Statistical comparisons between treatments were conducted using linear mixed-effects models, and no significant differences were found for any soil health indicators examined ($P > 0.05$). WAS is wet aggregate stability, POXC is permanganate oxidizable carbon, OM-LOI is organic matter loss-on-ignition, Min-C is mineralizable carbon, WEOC is water extractable organic carbon, TOC is total organic carbon, ACE-N is soil protein, TN is total nitrogen, STP is soil test phosphorus, STK is soil test potassium.

2020). A continuous no-tillage system can also lead to pH stratification due to fertilizer placement and reduced soil mixing, which can result in soil acidity close to the surface (Barth et al., 2018). For this reason, sampling the uppermost soil depth (0–5 cm) under no-tillage has been reported to be more effective for accurate lime recommendations in this system (Barth et al., 2018; Souza et al., 2023). Although soil pH was less in no-tillage compared to the conventional tillage system (Fig. 3), all no-tillage trials maintained an overall mean soil pH above the acid soil threshold of 5.5. Therefore, the observed low soil pH with no-tillage practices is unlikely to adversely affect nutrient availability for crop production in this region. Kibet et al. (2016) found no difference in soil pH (depth 0–10 cm) between no-tillage and conventional tillage in a study evaluating 33 years of tillage systems in corn and soybean rotation. In contrast, Li et al., 2020 in a meta-analysis using 240 data points, reported that soil pH in both topsoil (0–15 cm depth), and subsoil (>15 cm depth), decreased after 6 years of no-tillage compared to conventional tillage, and similar to what we found, pH average values for no-tillage systems remained above 5.5.

The linear mixed-effects models analysis revealed no statistically significant differences in soil health indicators between tile-drained and undrained treatments ($P > 0.05$; Fig. 4). This is intriguing since these studies have been managed in these systems for over 11 years. Thus, it is expected that greater soil water content in the undrained soil would reduce microbial activity, resulting in slower OM decomposition and more C accumulation (Fernández et al., 2017; Richardson et al., 2023). Contrary to our findings, other studies in the US Midwest have reported that undrained soils had greater TOC and TN (Fernández et al., 2017; Kumar et al., 2014) than drained soils. Abid and Lal (2008) found that

SOC was greater in no-tillage and undrained treatments than in drained conventional tillage treatments. The ability of artificial drainage to reduce excess water in the soil early in the season−particularly during wet years, which facilitates earlier planting−might be associated with the greater yields reported in drained fields compared to undrained fields across the North Central US region (Mourtzinis et al., 2021). While artificial drainage was not associated with changes in soil health indicators for corn-soybean and soybean-wheat crop systems, our study was limited to only two sites including no-tillage, reduced tillage and conventional tillage (total of 48 observations). Therefore, we encourage conducting additional studies exploring the effect of this practice on soil health indicators and crop yield across various soil types and climate zones.

4.2. Overall implications, study limitations, and future research

Our comprehensive analysis across 21 research trials provides valuable insights into the complex interplay between management practices and soil health in soybean-based cropping systems. While our hypothesis found partial support for cover crops (Min-C, WEOC indicators) and crop rotation (STP), and limited support for tillage (pH only) and drainage (no observed differences), the findings suggest that soil health improvement may be a gradual process and highly dependent on the indicator evaluated and the environmental context. The lack of response to management for the remaining soil tests assessed—particularly those related to total organic carbon and physical stability—illustrates the complexity of soil health dynamics in systems often characterized by limited rotation diversity and a history of intensive management (Dietz et al., 2024). Despite the cover crop trials averaging only five years, the association of cover crops with greater labile C indicators (Min-C and WEOC) highlights the potential of cover crops to enhance soil health. In addition to the many other benefits associated with cover crops, such as reduced soil and nutrient loss, weed suppression, and enhanced water quality (Chen et al., 2022; Osipitan et al., 2018), cover crops are a key component for maintaining or improving soil health.

These encouraging results support the adoption of cover crops by farmers aiming to build soil health through more sustainable practices. The adoption of cover crops, especially in low-residue crops such as soybean, could benefit the cropping system by offering the potential to add extra residue and immobilize and retain soluble nutrients for the subsequent crop (Hunter et al., 2019). However, it is important to keep in mind that the cover crop species adopted, weather conditions, and management practices play essential roles in the ability of the cover crop to produce enough biomass for soil health benefits without compromising crop yield.

It is important to acknowledge some limitations in this study. The collection of soil samples at a single time point during the 2023 growing season limited our ability to assess temporal changes in soil parameters. Additionally, we collected soil samples from the surface or arable layer (0–15 cm depth), which may not fully capture changes in SOC pools at deeper soil layers (Dietz et al., 2024; Olson and Al-Kaisi, 2015). Furthermore, the field trials did not always include the same background treatment; for example, cover crop trials were conducted under both no-tillage and reduced tillage systems. Despite these limitations, the use of meta-analysis enabled us to identify associations between soil health parameters and management practices in field trials with varying climates and soils. To address this study limitations, future research should focus on well-designed and common regional long-term experiments to explore the mechanisms behind soil health changes due to management practices, elucidating current knowledge gaps and informing farmers' management practices associated with soil health improvements and their impact on crop yield.

5. Conclusions

This study investigated whether management practices are associated with changes in recommended soil health indicators at the 0–15 cm depth in 4- to 50-year soybean cropping system trials across the US. Our findings partially support our hypothesis regarding conservation practices. The inclusion of cover crops was associated with greater mineralizable carbon (Min-C) and water extractable organic carbon (WEOC), indicating improved biological activity and labile carbon pools in the soil. Furthermore, two-crop rotations were associated with greater soil test phosphorus (STP) values. While no-tillage resulted in more acidic pH compared to conventional tillage, the remained soil health tests did not show changes in response to the assessed management practices. Similarly, our hypothesis regarding the negative impact of artificial drainage was not supported, as no significant differences were observed in soil tests between tile-drained and undrained treatments for any of the indicators examined. While cover crops show promise for enhancing specific biological indicators, the lack of responses across other parameters suggests that a combination of integrated long-term conservation practices, tailored to specific regional and soil conditions, may be necessary for a more comprehensive soil health assessment. This study provides valuable insights and a unique open-access dataset to inform future research and help farmers in making sustainable management decisions for soybean production.

CRediT authorship contribution statement

David Jordan: Writing – review & editing, Investigation. **Chad D. Lee:** Writing – review & editing, Investigation. **Matthew D. Ruark:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **John M. Gaska:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Jeremy Ross:** Writing – review & editing, Investigation. **Herman J. Kandel:** Writing – review & editing, Investigation. **Hanna J. Poffenbarger:** Writing – review & editing, Investigation. **Mark A. Licht:** Writing – review & editing, Investigation. **Lindsay Chamberlain Malone:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Maninder Pal Singh:** Writing – review & editing, Investigation. **Tatiane Severo Silva:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Laura E. Lindsey:** Writing – review & editing, Investigation. **Joseph G. Lauer:** Resources, Investigation. **Spyridon Mourtzinis:** Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation. **Rodrigo Werle:** Writing – review & editing, Investigation. **Rachel A. Vann:** Writing – review & editing, Investigation. **Michael Plumblee:** Writing – review & editing, Investigation. **Shawn P. Conley:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Trenton L. Roberts:** Writing – review & editing, Investigation. **Seth L. Naeve:** Writing – review & editing, Investigation.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Matthew Ruark reports financial support was provided by National Institute of Food and Agriculture. Shawn Conley reports financial support was provided by Wisconsin Soybean Marketing Board. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data analized for this study are available via Dryad https://doi.org/10.5061/dryad.v9s4mw787.

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